8Lab
Stochastic analysis: probabilistic/continuous-stochastic logic from model checking to simulation

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One slide sum-up on Stochastic Analysis

Concepts

- logics is the proper tool to capture system properties, their meaning, and their verification
- the transition from TS to DTMC and then to CTMC should be followed at the logic level
- this is: LTL/CTL, then PCTL, and then CSL
- in CSL we essentially estimate the overall probabilities of paths satisfying certain properties (U,X,F,G)

On verification

- model-checking: smart intelligent exploration of (finite) state space, exact result
- approximate model-checking: smart partial exploration of state space, with approximate result (ϵ, δ)
- simulation: draw of few runs/paths, especially for complex simulations

Tools

- need performance, hence, ad-hoc tools
- PRISM (symbolic modelchecker), will see Alchemist as a general simulator

Starting point and goals

References

- 08-repo: from virtuale (`https://github.com/mviroli/asmd-public-models`)
- PRISM, to install (`https://www.prismmodelchecker.org/`)

General goals

- get acquainted with using multiple simulation to analyse CTMCs
- get acquainted with PRISM model checker
- play with SPN in PRISM

Operational Steps

Step 1 – Download and check the repo

- Download the lab repository
- Import it in IntelliJ as an SBT project
- Get acquainted with all parts
  - `utils`: just the `MSet` abstraction, and also others
  - `modelling`: the actual API/DSL definition with `CTMC`, `CTMCSimulation`, `CTMCExperiment` (meta-meta-model), and then `SPN` and `DAP` (meta-model)
  - `examples`: extensional example `StochasticChannel`; intensional by Petri Nets `StochasticMutualExclusion` and by DAP `DAPGossip`
  - `test`: checks and tests
- Check that all Scala tests and checks pass and why, specifically checking `CTMCExperiment`

Step 2 – Download, install and run PRISM

- Download it and run it
- Play with the provided code, possibly directly on CTMCs
- Try to reproduce runs shown in slides 26 – 29
- Recall the progress: edit model, create property with parameters, run simulation, or experiment, or experiment with simulation

Step 3 – Compare PRISM and SCALA tools

- Take the communication channel example, and perform comparison of results between PRISM and our Scala approach

Tasks

PRISM

- Make the stochastic Readers & Writers Petri Net seen in lesson work: perform experiments to investigate the probability that something good happens within a bound
- Play with PRISM configuration to inspect steady-state proabilities of reading and writing (may need to play with options anche choose "linear equations method")

PRISM-VS-SCALA

- Write Scala support for performing additional experiments and comparisons (e.g., G formulas, steady-state computations)

LLM-STOCHASTIC-ANALYSIS

- PRISM is rather well known, and perhaps LLMs know it. Can LLMs understand the meaning of a stochastic property? Can they solve (very) simple modelchecking? Can they preview what a simulation can produce?